Project Report For CS661: BIG DATA VISUAL ANALYTICS
Group 3
2025-2026 Summer Term

F1 Visual Analytics — A Dashboard for F1 Performance, Strategy, and Circuit Intelligence

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1. Introduction:

Formula 1 produces an unusually large and structured stream of data every race weekend — qualifying times, grid positions, lap-by-lap timing, pit-stop durations, championship standings, and classified results across many circuits and constructors. This richness makes it a natural fit for a visual-analytics project: the raw data is relational and normalized, spread across many small tables, and answering even a simple question such as "does starting position determine the race result" requires joining and aggregating several of these tables before any chart can be drawn. The goal of this project was to take that raw, relational Formula 1 dataset and turn it into a single interactive dashboard that lets a user explore driver performance, constructor evolution, circuit characteristics, race strategy, and championship progression, without needing to touch a spreadsheet or write a query.

The underlying data was sourced from the Ergast Developer API , which contains Formula 1 records from 1950 onwards across multiple interconnected tables (races, results, drivers, constructors, circuits, qualifying, lap times, pit stops, sprint results, and championship standings). Since the objective was to analyze the current regulatory era of the sport, the dataset was restricted to the **2019–2024** seasons, giving **2,559 driver–race records** across **31 circuits** in **27 countries** and **15 constructors**. Rather than querying this relational structure at visualization time, the team built a preprocessing pipeline that denormalizes it into a single analytical table, one row per driver per race, enriched with a large set of engineered performance, reliability, and strategy metrics. Every dashboard module reads from this single cleaned and feature-engineered dataset, which keeps every chart in the application consistent with every other chart.

The final deliverable is a multi-page Dash (Plotly Dash) application with seven analytical modules: an Overview / Data Quality page, a Qualifying vs Race Analysis page, a Championship Evolution page, a Driver Performance Explorer, a Constructor Evolution module, a Circuit Intelligence module, and a Race Strategy Analysis module. Each module was designed around a specific analytical question rather than as a decorative chart, and each chart in the dashboard is accompanied by an explicit justification of why that chart type was chosen over the alternatives.

Objective

The project addresses the following concrete goals:

- Build a single, reliable, denormalized dataset from the raw relational Formula 1 tables (2019–2024), with documented cleaning and missing-value handling.
- Engineer a shared library of derived performance, reliability, strategy, and championship metrics so every module in the dashboard uses consistent definitions.
- Provide an at-a-glance overview of dataset scale, coverage, and data quality before any deeper analysis begins.
- Quantify how closely a driver's starting (qualifying) position predicts their final race result, and by how much drivers typically gain or lose positions on race day.
- Track championship progression race-by-race, including cumulative points and rank movement, rather than only the final standings.
- Provide a multidimensional, driver-level performance explorer that supports head-to-head and teammate comparisons.

- Analyze constructor-level trends: which teams improved, declined, dominated, or struggled with reliability across seasons.
- Analyze circuit-level characteristics: how track design and location relate to lap speed, overtaking, and reliability.
- Analyze race strategy: how pit-stop count, duration, and consistency relate to final race outcomes.
- Integrate all of the above into one cohesive, consistently styled, interactively filterable dashboard rather than a set of disconnected scripts.

Tools and Technologies Used-

- **Python** — primary language for the entire preprocessing pipeline and dashboard application.
- **Pandas** — data loading, cleaning, merging, aggregation, and feature engineering across all relational tables.
- **Dash** — the web framework used to build the multi-page, callback-driven interactive dashboard.
- **Plotly / Plotly Express** — charting library used for every visualization in the dashboard (scatter, line, bar, heatmap, area, choropleth/map, radar, box, violin, bubble, and dumbbell-style charts).
- **dash-bootstrap-components** — component library used for layout, theming (CYBORG dark theme), tabs, and styled dropdowns/sliders.
- **dcc.Store** — Dash component used to persist shared application state (theme choice, active filters) across page navigation.
- **Leaflet (via Dash)** — interactive geographic mapping for the circuit distribution view.
- **Custom CSS** — a shared design-token system giving the dashboard a consistent Formula 1-inspired visual identity across all seven modules.

2. Tasks:

The project was scoped as seven concrete analytical tasks, each owned by a subset of the team and each mapped to one dashboard module:

1. Cleaned and integrated the raw relational Formula 1 datasets into a single analytical table, summarized dataset scale and quality on an Overview page, and engineered derived performance metrics to analyze the relationship between qualifying position and race results (Shreyansh Singh & Daksh Singh).
2. Build the dashboard shell, global filters, and the cumulative championship-points and rank-movement views (Nishant Singh, Harsh Maheshwari).
3. Build a multidimensional driver performance explorer supporting progression, consistency, and multi-metric comparison (Pruthviraj Meghe).

4. Analyze constructor-level improvement, dominance, and reliability across seasons (Snehasish Haldar).
5. Analyze circuit-level speed, overtaking, and reliability patterns, including a geographic view (Pratik Kumar).
6. Analyze how pit-stop strategy relates to final race outcomes (Vishal Kumar).

3. Proposed Solution:

0.1 Data Sources and Preprocessing

The complete Ergast dataset contains Formula 1 records from 1950 onwards. Since the objective of this project is to analyze modern Formula 1 racing, the dataset was restricted to the **2019–2024** seasons. This filtering was performed on `races.csv` by selecting only races occurring in that window; the resulting `raceIds` were then used to filter every race-dependent table, including Results, Qualifying, Driver Standings, Constructor Standings, Lap Times, Pit Stops, and Sprint Results. Reference tables such as Drivers, Constructors, Circuits, and Status were retained in their entirety, since they supply metadata required during the merge.

Cleaning- Before merging, the pipeline removed records outside the selected seasons, verified primary/foreign key consistency, converted date columns to `datetime`, standardized column names to avoid duplicate identifiers after merging, and validated unique identifiers across every table. No duplicate records were found after filtering.

Aggregating high-granularity tables- Two source tables were far finer-grained than needed for visualization. `lap_times.csv` contains one row per driver per lap; this was aggregated to one record per `(raceId, driverId)` pair, computing total laps completed, average lap time, fastest lap, slowest lap, and the standard deviation of lap times. `pit_stops.csv` similarly contains one row per pit stop and was aggregated to `(raceId, driverId)`, computing total pit-stop count, average duration, and fastest/slowest stop.

Master dataset construction- The individual cleaned and aggregated tables — race results, race metadata, driver information, constructor information, circuit information, race status, qualifying results, driver and constructor championship standings, aggregated lap statistics, aggregated pit-stop statistics, and sprint results — were merged into a single denormalized table at **driver–race granularity**: each row represents the complete performance of one driver in one race. This eliminates repeated joins at visualization time and lets every dashboard module read from one dataset.

Missing-value handling- Several columns naturally contain missing values because of how Formula 1 weekends are structured — sprint columns are empty for non-sprint weekends, Q2/Q3 times are unavailable for drivers eliminated earlier in qualifying, pit-stop statistics are empty for drivers who never stopped, and certain timing variables are unavailable for retired drivers. These were treated as meaningful missingness (a real racing condition) rather than a data-quality defect and were intentionally retained rather than imputed or dropped.

Validation- After merging, the pipeline verified that every row corresponds to exactly one driver in one race, confirmed no duplicate `(raceId, driverId)` pairs existed, validated row counts against the original results table, inspected missingness across all columns, and confirmed that lap and pit-stop aggregates aligned correctly with race results.

0.2 Feature Engineering

On top of the raw merged columns, the pipeline derives a shared library of analytical features consumed across every dashboard module:

- **Performance features** — positions gained (grid minus finish), winner/podium/points-finish flags, and average finish/grid/points per driver.
- **Reliability features** — a `is_dnf` indicator (only "Finished" and "+n Laps" statuses count as classified finishes; everything else — accidents, mechanical failures, disqualifications — is treated as a DNF), plus constructor reliability percentage, win rate, and average finishing position.
- **Strategy and pace features** — pit-stop participation, lap consistency score, race-pace ranking, and deltas from the fastest and average race pace.
- **Championship features** — gap to the championship leader (driver and constructor), leader indicator, and title-contender indicator.
- **Driver features** — age on race day and years of experience within the dataset window.
- **Qualifying features** — Q2/Q3 qualification flags and qualifying-to-race position improvement.
- **Sprint and temporal features** — sprint participation indicator and race month.

The resulting integrated table, `master_f1_final.csv` (also referenced as `master_f1_final_dnf_fixe` and `derived_f1_metrics.csv` by specific modules), is the single source of truth consumed by every page of the dashboard.

0.3 Application / Dashboard Architecture

The dashboard is a Dash (Plotly Dash) single-page application (SPA): page navigation swaps content dynamically without a full reload. Its architecture has three layers:

- **Data layer-** A centralized `load_data()` function performs safe ingestion (catching missing/corrupted files), numeric cleaning (columns such as `year`, `grid`, `positionOrder`, `points`, and `championship_points`), and shared feature engineering (pit-stop duration in seconds, lap consistency score, valid-pit-stop indicator, race label). Every one of the seven modules consumes the output of this single pipeline, which is what keeps results consistent across the whole application.
- **Shell / integration layer-** A responsive sidebar navigation, routing logic, and reusable page-layout system implement the SPA pattern. A global CSS design system (custom color palette, typography, responsive grid, hover animations) gives every page a consistent Formula 1-inspired visual identity, and `dcc.Store` persists the user's theme choice and active filters as they move between pages instead of resetting on every navigation.
- **Module layer-** Each of the seven analytical modules is a self-contained page with its own filters and callbacks, but all share the same underlying dataset, the same styling function for Plotly figures (dark template, consistent font, transparent background, matching gridlines), and the same interaction conventions (dropdowns for categorical filters, range sliders for season spans).

Integrating seven independently developed modules into one cohesive application required resolving callback conflicts, standardizing callback dependencies, and validating that every module behaved correctly once combined with the others, as well as performance work such as avoiding unnecessary data reloading and wrapping heavier visualizations in loading indicators.

4. Results:

This section documents each of the seven dashboard modules using the Task Overview / Information Represented / Observed Trends / Inference and Insights / Real-World Implications structure, in the order the modules appear in the dashboard's development and navigation sequence.

0.4 Overview Page

0.4.1 Task Overview

The Overview page is the dashboard's entry point. Its purpose is to prove, before any deeper analysis begins, that the underlying dataset is complete, clean, and ready for analysis, and to orient a new user to its scale.

0.4.2 Information Represented

KPI cards summarizing dataset scale (total records, seasons, drivers, constructors, circuits); a missing-value heatmap across all columns; a season-coverage bar chart; and a dataset preview table.

0.4.3 Observed Trends

The dataset contains 2,559 driver–race records across the 2019–2024 seasons, 31 circuits, and 27 countries. Missingness is concentrated in a small number of expected columns — sprint-related fields, Q2/Q3 times, and pit-stop statistics — rather than being scattered randomly across the table, which is consistent with those gaps representing genuine racing conditions (no sprint that weekend, eliminated in Q1, no pit stop taken) rather than data-quality problems.

0.4.4 Inference and Insights

Because missingness clusters in explainable columns rather than appearing randomly, the team chose to retain rather than impute these values — an imputed Q3 time for a driver eliminated in Q1 would be actively misleading. The heatmap format was chosen specifically because it exposes this pattern at a glance, in a way a plain table of missing-value counts would not.

0.4.5 Real-World Implications

A visible, trustworthy data-quality summary is what lets every other module in the dashboard be taken at face value; an analyst opening the driver performance or race-strategy pages does not have to separately audit missingness before trusting a chart's KPI numbers.

0.5 Qualifying vs Race Analysis

0.5.1 Task Overview

This module quantifies how strongly a driver's starting grid position predicts their final race result, and by how much drivers typically gain or lose positions once the race begins.

0.5.2 Information Represented

A jittered scatter plot of grid position vs. final finish position with a diagonal reference line, split by constructor; a 2D density heatmap over the same grid/finish space; and a Pearson correlation matrix between grid position, finish position, and positions gained. A MODE toggle (Compare Teams vs. Isolate One), a constructor selector, and a season-span range slider drive all three charts and three stat tiles through a single primary callback.

Figure 1: Grid position vs. final finish position, filtered to Red Bull, Mercedes, and Ferrari, 2019 season

0.5.3 Observed Trends

For the sampled 2019 Red Bull / Mercedes / Ferrari selection (124 results), the average change in position was −0.64, meaning drivers in this window tended to lose slightly more positions than they gained. The single largest gain was Vettel's +18 positions in one race. The density heatmap shows the brightest concentration of results between grid positions 1–3 finishing in positions 1–3, with concentration dropping sharply past roughly grid position 6–7. The correlation matrix reports grid–finish correlation of 0.33, positions-gained–finish correlation of −0.69, and positions-gained–grid correlation of 0.46.

0.5.4 Inference and Insights

Front-row starts convert into strong finishes far more reliably than any other outcome, but the moderate (0.33) grid–finish correlation shows starting position influences, without determining, the final result. Drivers starting further back have more room, on average, to gain places ($r = 0.46$ against grid), while Vettel's +18 outlier demonstrates that a poor grid slot can still be overturned on race day.

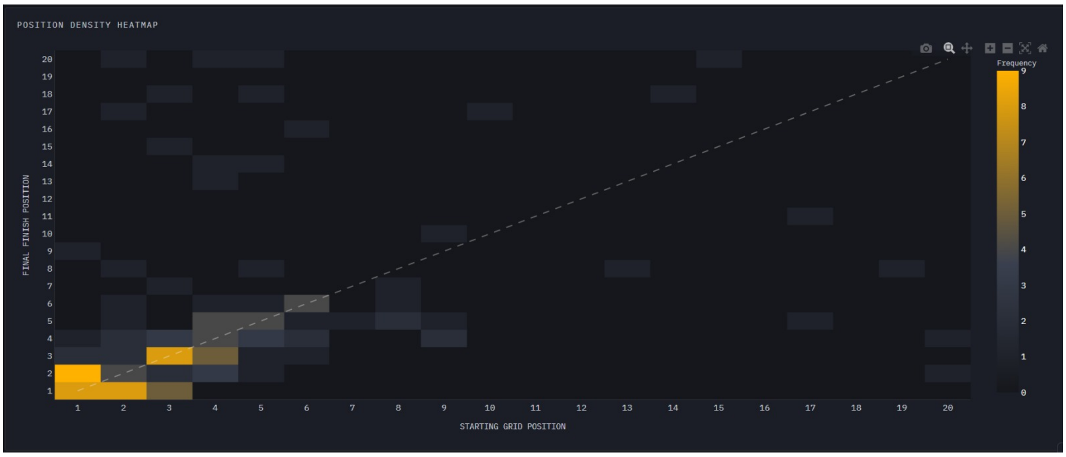

Figure 2: Frequency of each grid-to-finish combination for the same filtered selection.

0.5.5 Real-World Implications

This view is directly useful to team strategists and broadcasters framing a race weekend: it quantifies exactly how much a qualifying result "matters" for a given team or era of car, rather than relying on intuition.

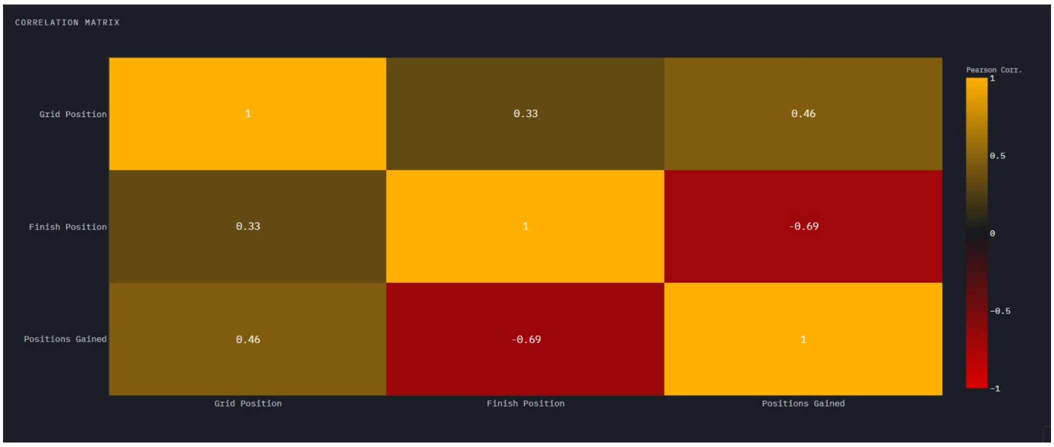

Figure 3: Pearson correlation between grid position, finish position, and positions gained.

0.6 Championship Evolution

0.6.1 Task Overview

This module tracks championship progression across a season, going beyond final standings to show how title battles evolved race by race.

Figure 4: Global circuit distribution map (Leaflet)

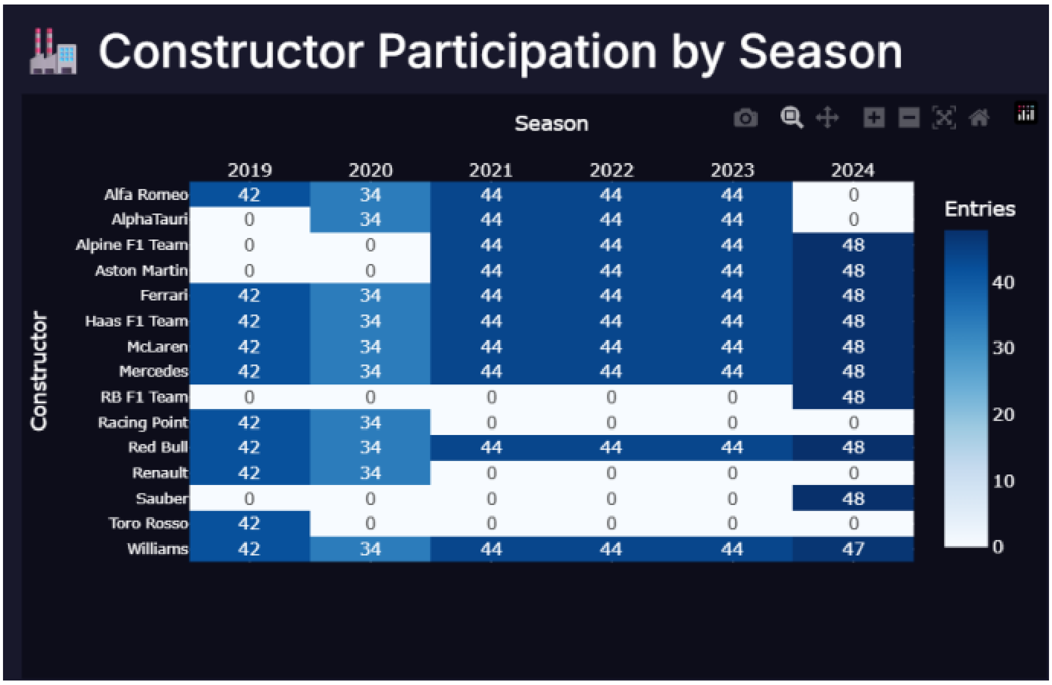

Figure 5: Constructor participation Heatmap

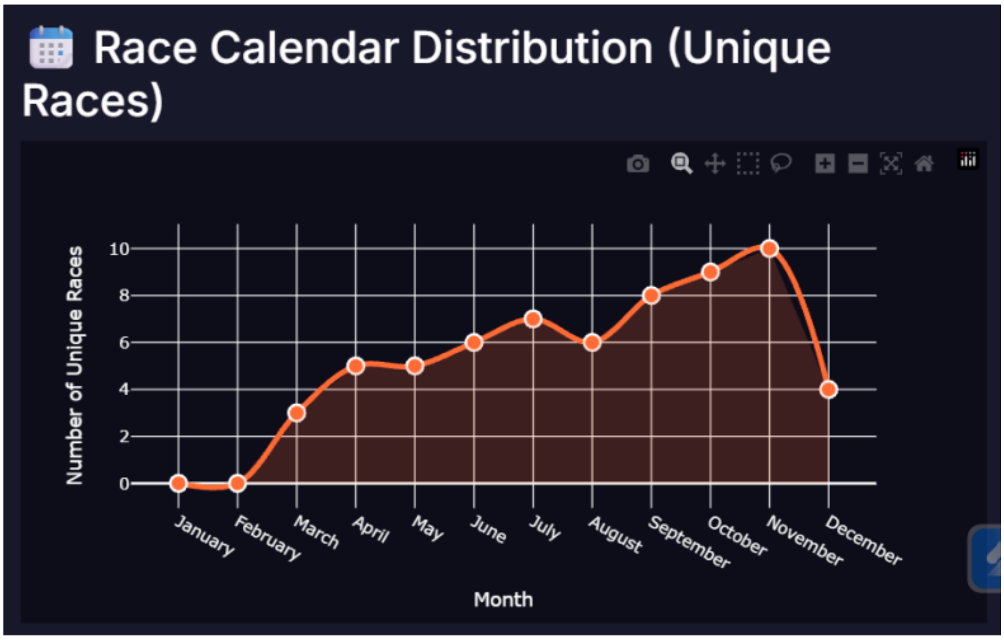

Figure 6: Race Calendar Seasonality

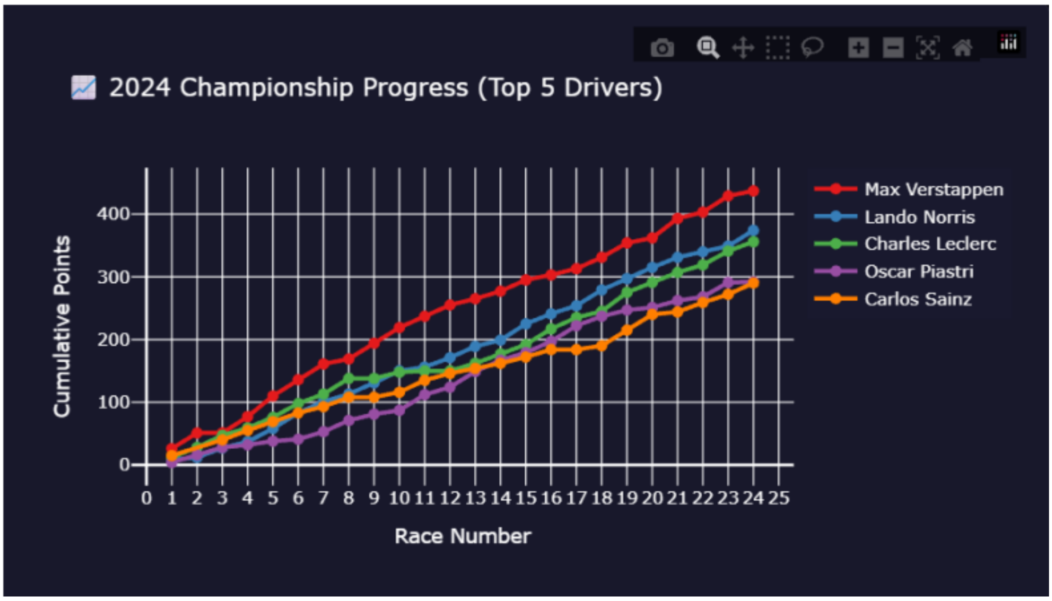

Figure 7: Ranked standings table paired with a cumulative line chart

0.6.2 Information Represented

A global circuit distribution map; a constructor participation heatmap across seasons; a race-calendar seasonality view; a cumulative-points line chart paired with a ranked standings table for a selected season; a championship bump/rank-movement chart; and a historical champions timeline with auto-generated season insights.

0.6.3 Observed Trends

The cumulative-points line chart shows accumulating championship totals race-by-race for the leading drivers in a selected season, making the shape of a title fight (a close battle vs. a runaway lead) visible directly, rather than only its final outcome. The bump chart shows rank position changing round by round, surfacing momentum shifts that a single end-of-season table cannot.

0.6.4 Inference and Insights

A line chart was chosen for cumulative points because championship progress is inherently ordered round-by-round, and a bar chart would clutter rather than clarify momentum. A bump chart was chosen specifically for rank movement because it is designed to show change in rank over time, which a table of positions cannot convey visually.

0.6.5 Real-World Implications

This is the module most directly aimed at fans and commentators: it turns a season's worth of results into a legible narrative of how a championship was actually won or lost, not just who won it.

0.7 Driver Performance Explorer

0.7.1 Task Overview

This module treats driver performance as inherently multidimensional — points, finishing position, grid position, positions gained, and race-to-race consistency all at once — and lets a user track a driver's progression, compare them against a teammate or rival, and read their overall profile at a glance.

0.7.2 Information Represented

A filled line chart of finishing classification over time (reversed axis, so higher on the chart is better), with an optional second driver overlaid; a box/violin toggleable distribution chart of classified (non-DNF) finishing positions; and a five-axis radar chart (Qualifying Pace, Race Craft, Overtaking Efficiency, Scoring Capacity, Lap Consistency), each metric min–max normalized to 0–100 with grid and finish position inverted so that "further from center" always means "better" on every axis. A Target Focus Driver dropdown, an optional Compare Driver dropdown, a season filter, and a distribution-view toggle drive all charts through one primary callback.

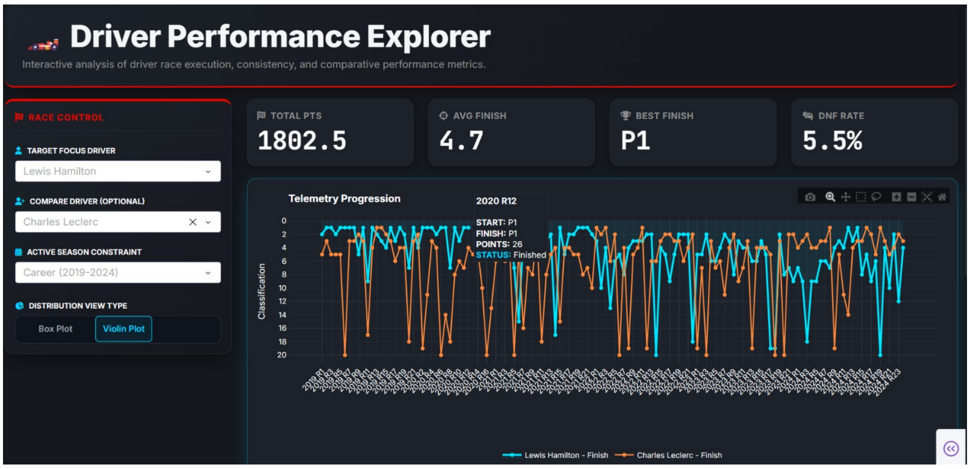

Figure 8: Telemetry Progression line chart — Hamilton vs. Leclerc, career 2019–2024.

0.7.3 Observed Trends

For a sampled Hamilton vs. Leclerc career comparison (2019–2024), Hamilton recorded 1,802.5 total points, a 4.7 average finish, a best finish of P1, and a 5.5% DNF rate. Hamilton's finishing-position violin is widest and tallest near P1–P4, while Leclerc's is widest around P4–P6 with a longer tail toward P16; Leclerc's DNF rate (14%, 18/128) is more than double Hamilton's (6%, 7/127) over the same window. The box-plot view shows Hamilton's interquartile range spanning roughly P2–P6 (median ≈ P3) against Leclerc's P3–P6 (median ≈ P4).

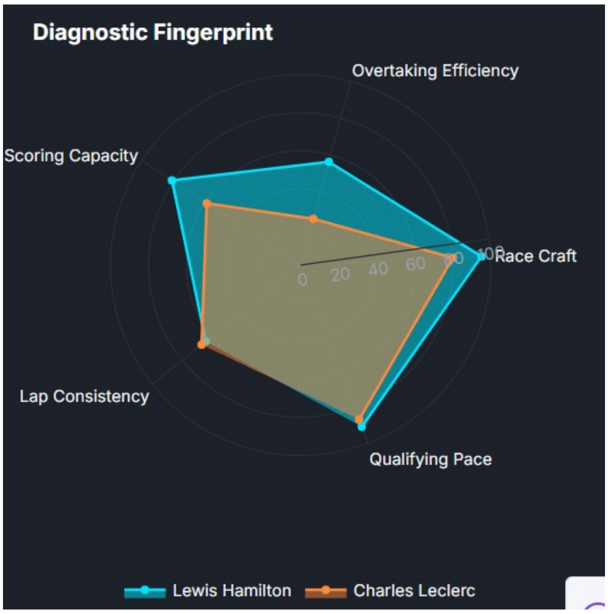

Figure 9: Radar Chart — Hamilton vs. Leclerc, career 2019–2024

0.7.4 Inference and Insights

A line chart was used for progression because it is the natural representation of an ordered sequence of races; a box/violin toggle was offered specifically because a single average-finish number would collapse two very different careers into one figure and hide consistency differences. The radar chart was deliberately kept secondary and capped at two drivers, since overlapping radar shapes become unreadable beyond that. Read together, the charts show Hamilton holding both a reliability edge and a tighter, higher finishing distribution than Leclerc over the sampled window.

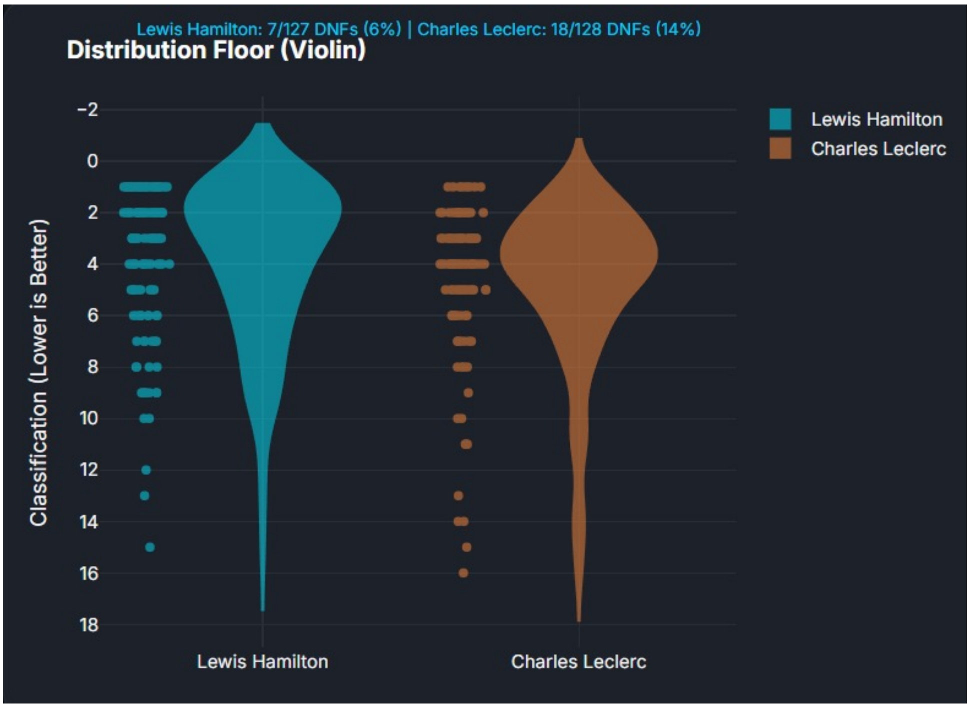

Figure 10: Violin Plot — Hamilton vs. Leclerc, career 2019–2024

0.7.5 Real-World Implications

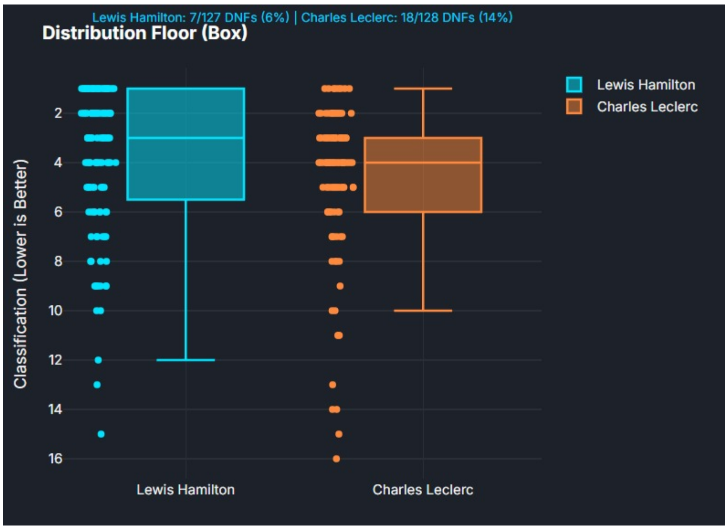

Figure 11: Box Plot — Hamilton vs. Leclerc, career 2019–2024.

This module mirrors how team principals and pundits actually evaluate drivers — not by one number, but by progression, consistency, and a compact multi-metric profile — making it directly transferable to scouting and contract-renewal discussions.

0.8 Constructor Evolution

0.8.1 Task Overview

This module explains how each constructor (team) improved, declined, dominated, or struggled with reliability across the 2019–2024 seasons, across 15 constructors (4 shown by default: Mercedes, Red Bull, Ferrari, McLaren).

0.8.2 Information Represented

A metric-switchable line chart (Season Points / Cumulative Points / Points Share %) with click-to-drilldown; a faceted Finish-vs-DNF stacked bar chart per constructor; a stacked points-share area chart; a linked drilldown view (DNF-cause table and donut chart) reachable by clicking a point on the line chart; live KPI cards (Total Points, Most Dominant, Best Improver, Most Reliable) with auto-generated insight sentences; and two sortable/filterable data tables (all-time leaderboard, year-over-year change). A constructor multi-select, a season range slider, and a metric radio drive every chart through a single callback.

Figure 12: Constructor Points by Season — Season Points view, 2019–2024

0.8.3 Observed Trends

In the Season Points view, Mercedes leads in 2019 (∼740 pts) and declines to the mid-300s/low-400s by 2023–2024, while Red Bull dips in 2020 and climbs to a 2023 peak near 795 points. McLaren closes from under 150 points (2019, 2022) to roughly 600+ points in 2024, the steepest single-season climb of the four. The cumulative view shows Red Bull overtaking Mercedes by 2024 (both around 3,300–3,400 cumulative points), with Ferrari a clear third and McLaren lowest overall despite its strong 2024. Points share shows Mercedes falling from roughly a third of all points in 2019 to under 20% by 2023, while Red Bull's share rises from under 20% to roughly a third over the same window. The reliability (Finish vs. DNF) chart shows all four constructors with a visibly worse 2020, and McLaren with a fully green (zero-DNF) 2024. A drilldown example (Ferrari, 2020) shows six DNFs split across five distinct causes rather than one dominant failure mode.

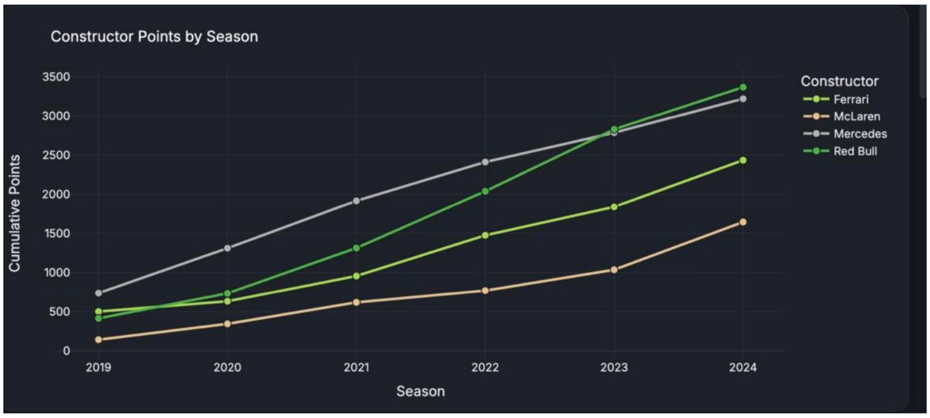

Figure 13: Cumulative Points view — running total, 2019 through 2024

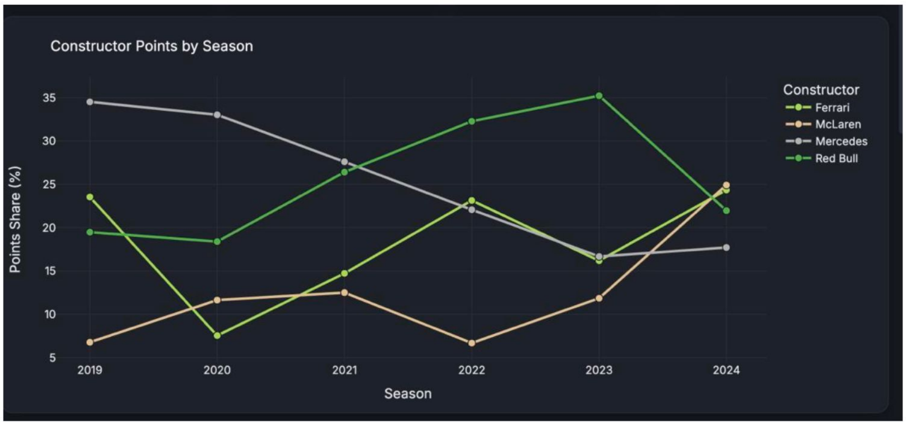

Figure 14: Points Share % view — dominance area chart

0.8.4 Inference and Insights

Three separate metric views (season, cumulative, share) were kept as one switchable chart rather than three separate ones so a viewer can compare trend shapes without re-learning the chart; a stacked bar was chosen for Finish/DNF composition because the split is categorical within a season, and an area chart was chosen over a pie chart specifically because it expresses dominance share continuously across time rather than as a single static snapshot. Read together, dominance changed hands mid-window (Mercedes → Red Bull), competitiveness and reliability are separable stories (Red Bull's strongest points seasons still carried DNFs; Mercedes kept strong finish rates even as its points fell), and the click-to-drilldown pattern reveals that a season's DNFs are rarely concentrated in one subsystem.

Figure 15: Finish vs. DNF composition, faceted by constructor, seasons 2019–2024

Figure 16: Drilldown example — DNF causes for Ferrari, 2020

0.8.5 Real-World Implications

This module is directly usable by team management and sponsors tracking a constructor's trajectory season over season, and by reliability engineers looking for whether retirements cluster by cause or era.

0.9 Circuit Intelligence

0.9.1 Task Overview

This module asks how circuit design and location relate to lap speed, overtaking, and reliability, across 31 circuits in 27 countries.

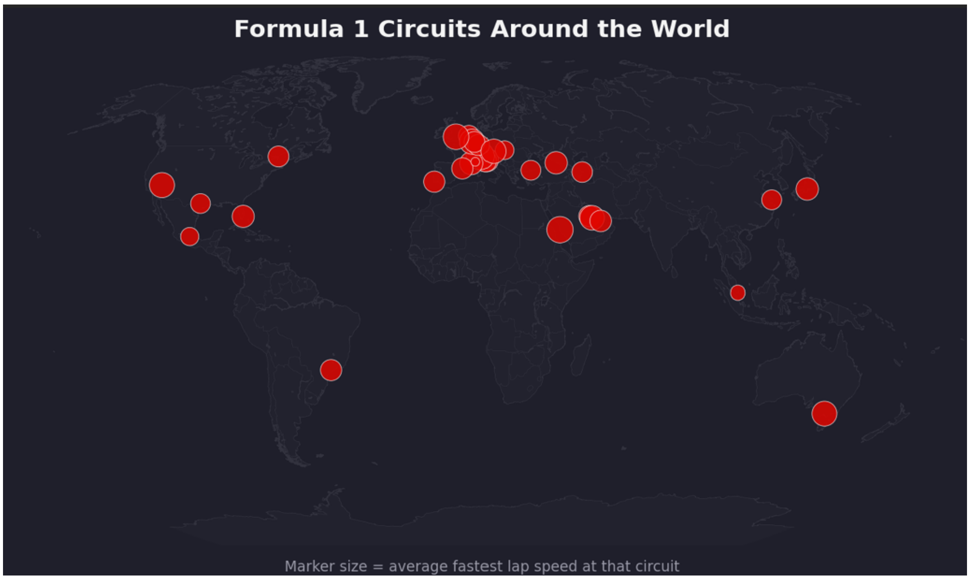

Figure 17: Formula 1 circuits across the world, with marker size representing average fastest lap speed

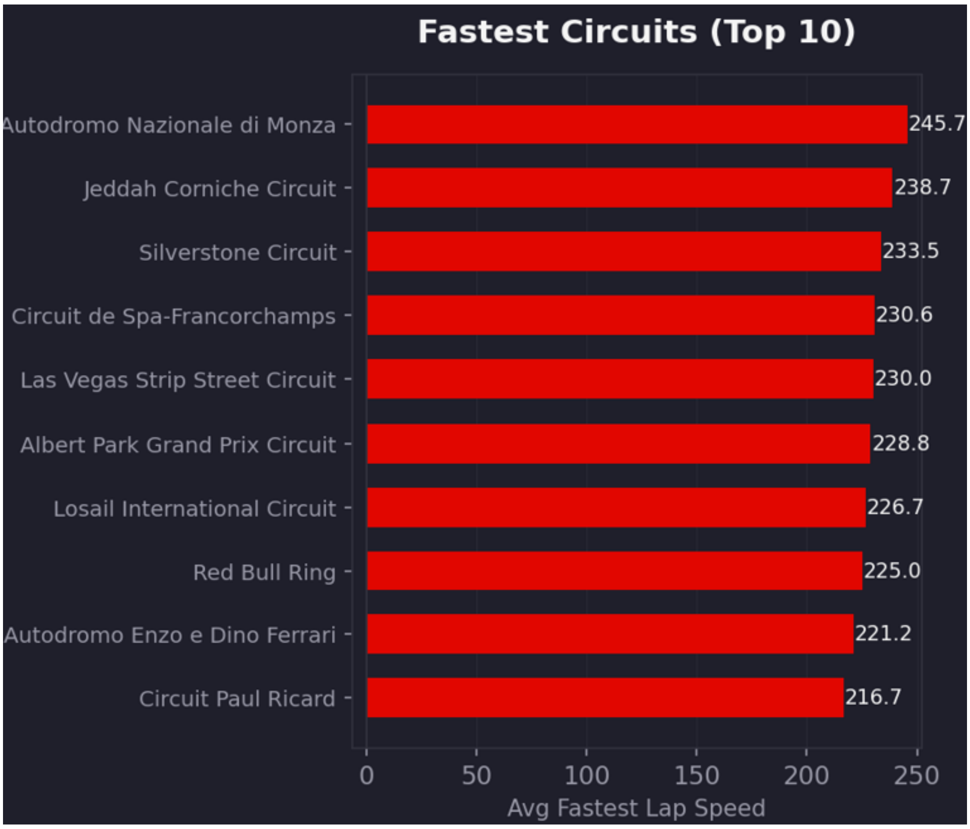

Figure 18: Top 10 Formula 1 circuits ranked by average fastest lap speed.

0.9.2 Information Represented

A geographic scatter map (marker size = average fastest lap speed); ranked bar charts of fastest circuits, position-change by circuit, and DNF rate by circuit (Top N / Bottom N configurable); a normalized, color-consistent heatmap of all three metrics together; and a bubble chart placing every circuit by speed (x) and position gain (y), with bubble size/color encoding DNF rate.

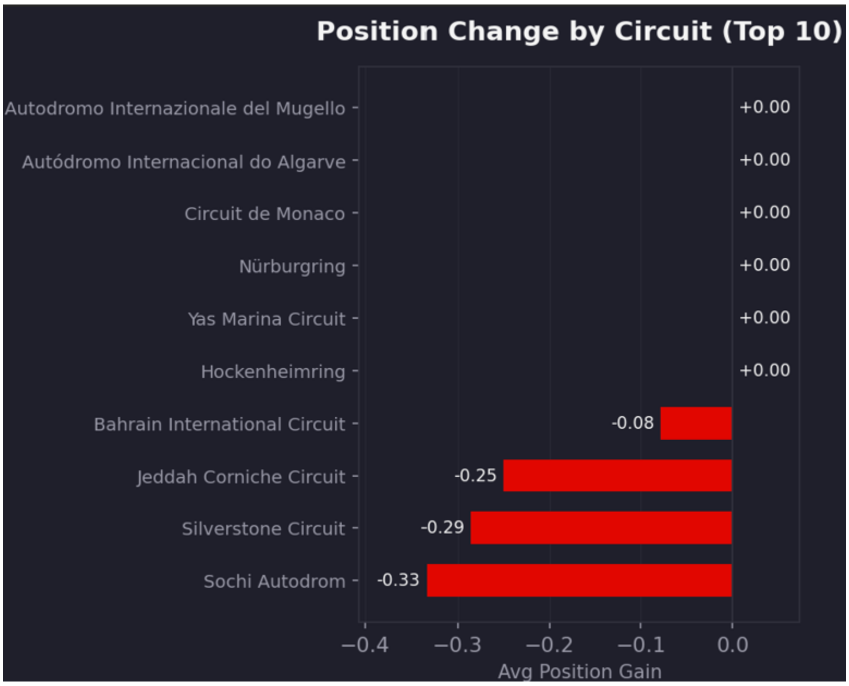

Figure 19: Top 10 Formula 1 circuits ranked by average position change during races

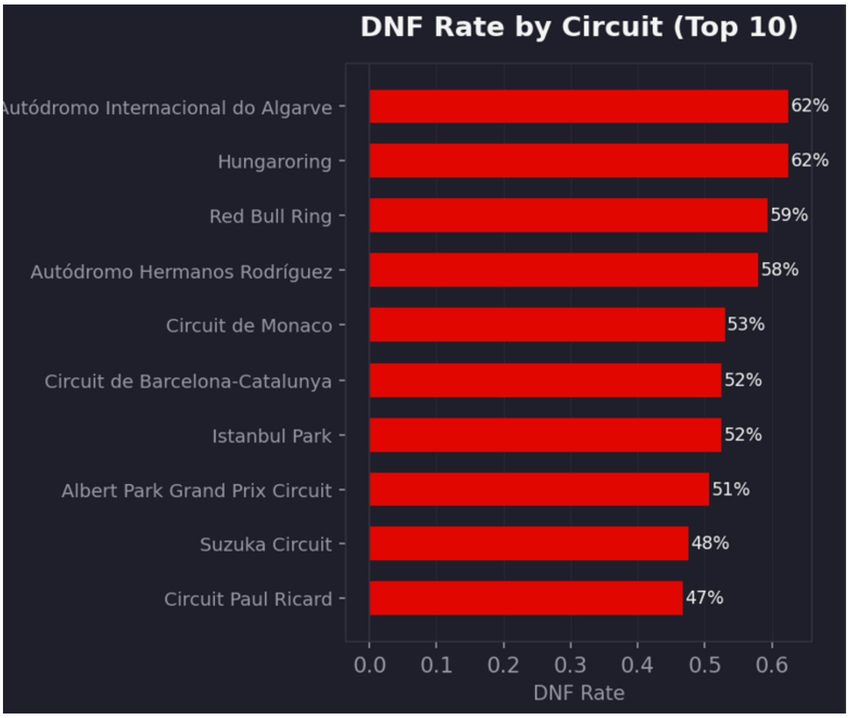

Figure 20: Top 10 Formula 1 circuits ranked by DNF rate

0.9.3 Observed Trends

Monza is the fastest circuit at 245.7 km/h average fastest lap speed, followed by Jeddah (238.7 km/h) and Silverstone (233.5 km/h); Monaco is slowest at 156.9 km/h. The mean across all 31 circuits is 209.9 km/h (median 206.7, std. dev. 18.2). Six circuits (Mugello, Algarve, Monaco, Nürburgring, Yas Marina, Hockenheimring) show an average position gain of exactly 0.000; the largest average losses are at Baku (−1.35), COTA (−1.14), and Zandvoort (−0.99); no circuit shows a positive average. Algarve and Hungaroring share the highest flagged DNF rate (62.5%); Miami (13.3%) and Las Vegas (17.5%) are the most reliable by this metric. Computed at the circuit level, speed correlates with DNF rate at $r = -0.36$; position gain is essentially uncorrelated with both speed ($r = 0.04$) and DNF rate ($r = 0.01$).

0.9.4 Inference and Insights

The ∼89 km/h Monza–Monaco gap is the widest of any pair of circuits and lines up with well-known layout differences, serving as a sanity check that circuit design, not season-to-season car development, dominates average-speed variation. A circuit average position gain of exactly zero is not missing data — it reflects that gains and losses mechanically cancel out within a fixed-size grid, which is why the few circuits with a consistent negative average (Baku, COTA, Zandvoort) are the interesting cases. The flagged DNF rate (42.4% dataset-wide) is roughly three times the true retirement rate (14.6%) because it also counts classified-but-lapped finishers, so the relative ranking of circuits is informative while the absolute percentages should not be quoted as literal retirement probabilities. The heatmap and bubble chart agree that reliability, not overtaking, is the metric worth cross-referencing against speed.

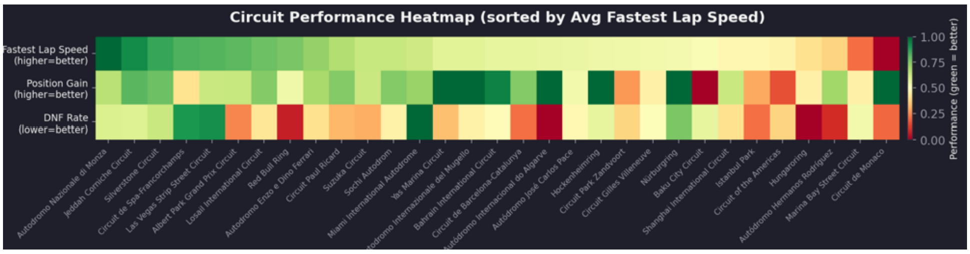

Figure 21: Circuit performance heatmap comparing speed, position change, and DNF rate

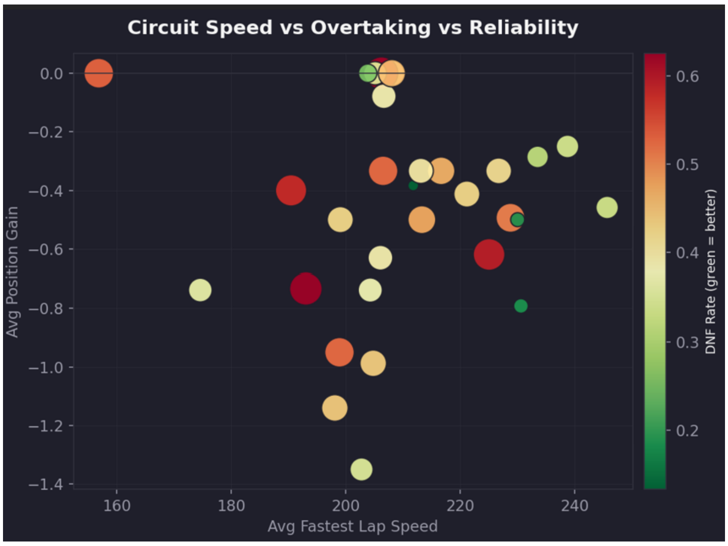

Figure 22: Circuit comparison based on average fastest lap speed, position change, and DNF rate

0.9.5 Real-World Implications

This module is directly useful to race engineers and strategists preparing for a specific circuit (expected overtaking difficulty, reliability risk) and to fans or broadcasters explaining why certain tracks are historically described as "street circuits" or "power circuits."

0.10 Race Strategy Analysis

0.10.1 Task Overview

This module asks how pit-stop strategy — stop count, duration, and team consistency — relates to final race outcomes, using 2,090 records that pass the module's normal-stop quality filters (of 2,559 total; 156 have zero recorded stops, 313 have long stops over 60 s, and 69 have grid = 0 pit-lane starts, each excluded by default).

0.10.2 Information Represented

A 2D density heatmap of pit-stop duration vs. positions gained with a mean trend line overlaid; a per-constructor box plot of pit-stop duration; a histogram of pit-stop duration distribution; a per-race dumbbell (grid → finish) chart colored by outcome with marker size encoding pit-stop count; and a strategy–outcome Pearson correlation matrix.

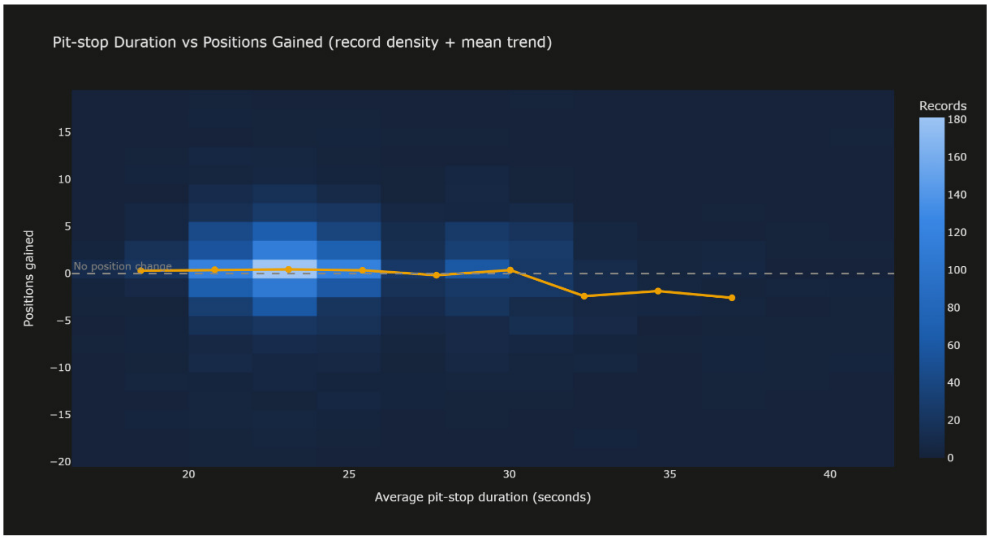

Figure 23: Where driver–race records concentrate (brighter = more records) with the average positions gained per pit-duration band overlaid in amber.

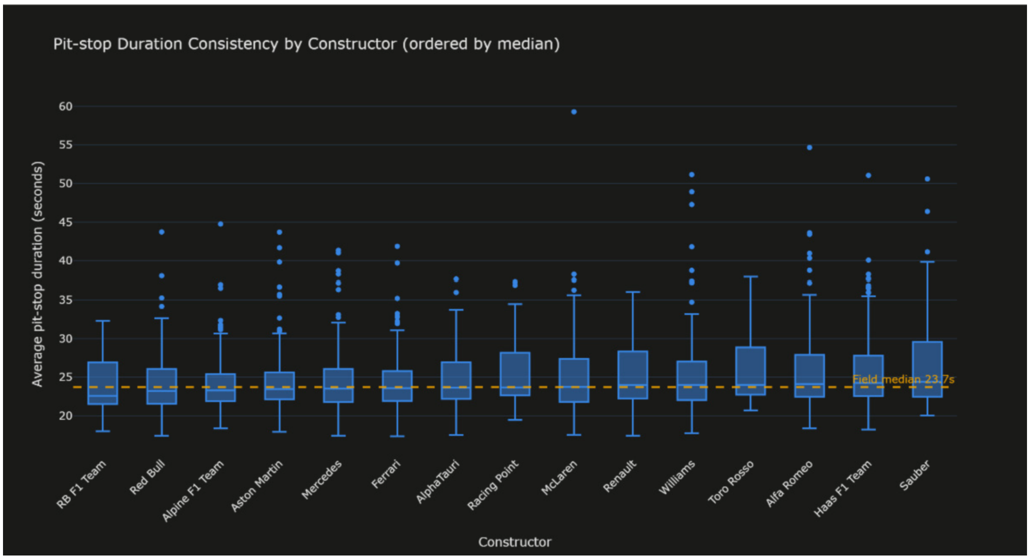

Figure 24: Distribution of average pit-stop duration for each team, ordered fastest-median first. The dashed line is the field median

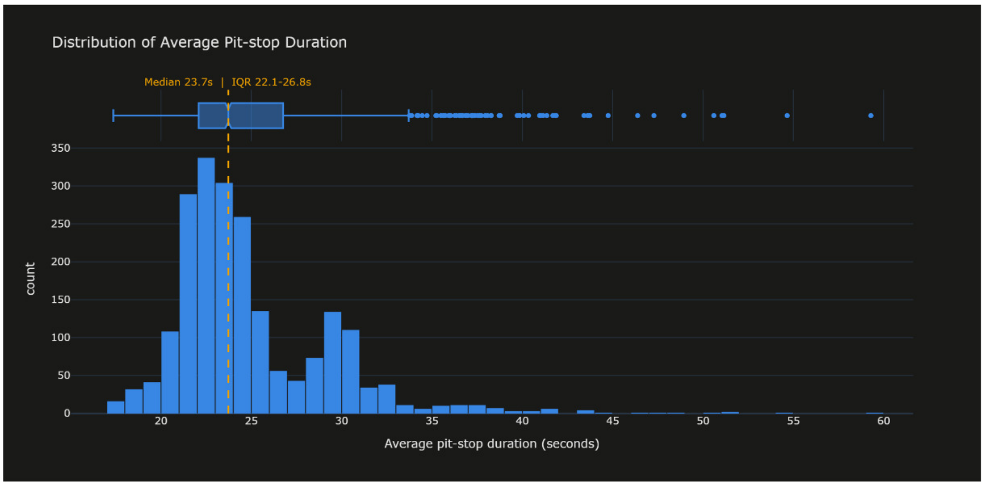

Figure 25: How often each pit-stop duration occurs across all records. The dashed line marks the median; the box on top summarises quartiles and outliers.

0.10.3 Observed Trends

The density heatmap shows the bulk of records between 20–27 seconds and around zero positions gained; the mean trend is roughly flat near zero up to ∼30 s, then drops to about −3 for the longest stops (mean gain +0.38 under 30 s vs. −1.07 at 30 s or more). Per-constructor medians are tightly bunched, from RB F1 Team (22.60 s, fastest median) to Sauber (24.41 s, slowest) — under a two-second spread — while spread and outlier counts vary far more between teams. The duration histogram is right-skewed and bimodal: a main peak at 22–24 s, a second bump around 29–30 s, and a long tail past 40 s (median 23.72 s, IQR 22.09–26.75 s). The correlation matrix shows strong, expected relationships as a sanity check (final position vs. points = −0.86; pace rank vs. final position = 0.83) alongside near-zero pit-strategy-to-outcome links (average pit duration vs. positions gained = −0.11; pit-stop count vs. positions gained = −0.04).

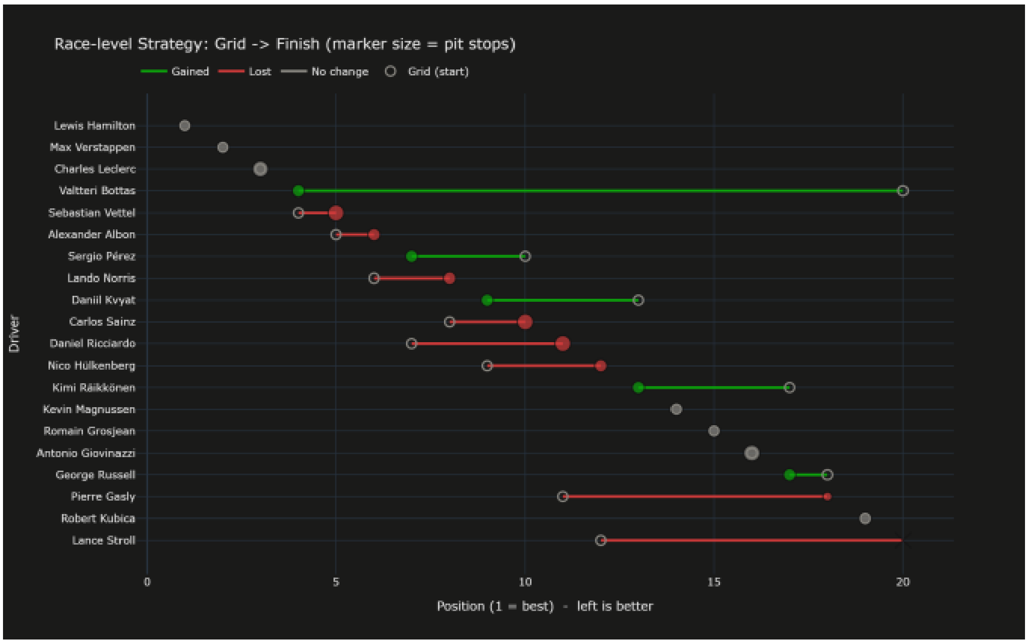

Figure 26: One race (2019 Abu Dhabi GP). Each driver's move from start (hollow marker) to finish (filled) is a connected line: green = gained, red = lost; marker size = number of pit stops.

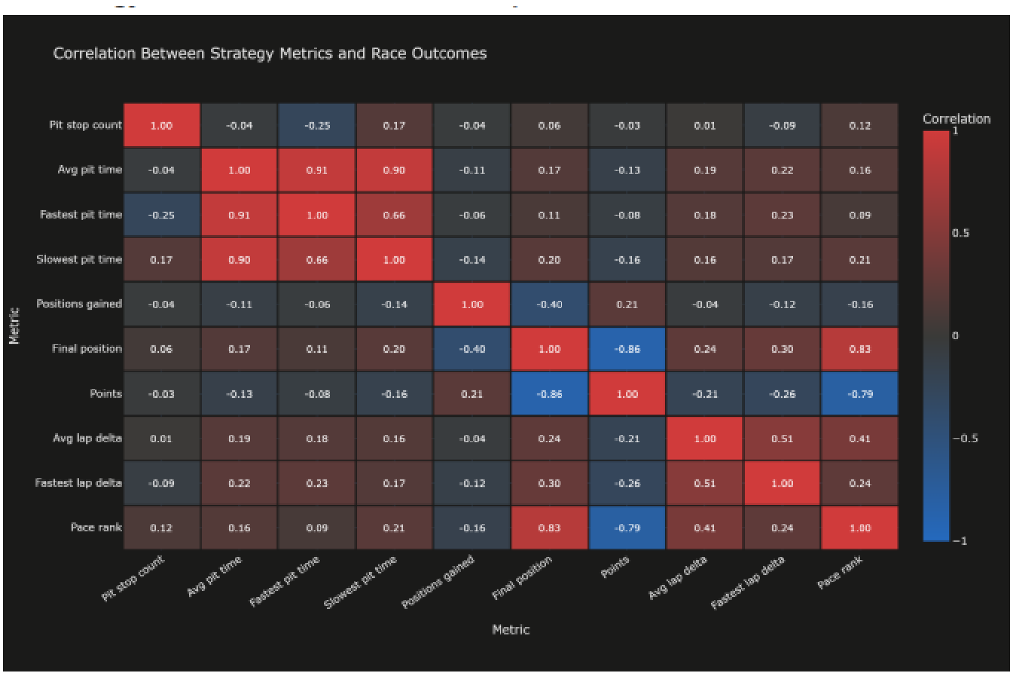

Figure 27: Correlation between every pair of strategy and outcome metrics. Red = positive, blue = negative, grey = near zero. The value is printed in each cell.

0.10.4 Inference and Insights

Faster pit stops, on their own, do not buy positions — being marginally quicker than average shows essentially no benefit — but a genuinely slow stop (30 s+) does cost roughly 1.07 places on average, so pit duration mostly functions as a penalty-avoidance factor rather than a way to gain ground. Team differences are concentrated in consistency (tight boxes, few outliers) rather than typical stop speed. Outcomes are driven far more by race pace than by pit execution, a conclusion the trend line and the correlation matrix independently agree on. The dumbbell view shows that individual drivers can still gain or lose many places within a single race, so while pit speed is a weak lever on average, race-specific circumstances create the large individual swings that the aggregate charts average out.

0.10.5 Real-World Implications

This is directly relevant to race strategists deciding how much value to place on marginal pit-crew speed gains versus avoiding a slow stop, and supports the broader finding that car/driver pace, not pit-lane execution, is the dominant lever on race outcome.

Challenges and Limitations

- **Relational-to-analytical transformation.** The source data's normalized, multi-table structure meant even simple questions required several joins; building and validating a single reliable denormalized table (avoiding duplicate `(raceId, driverId)` rows, preserving key integrity) was a real engineering task before any chart could be built.

- **Metric ambiguity in the DNF flag.** The dataset's `is_dnf` indicator counts any non-"Finished" status, including drivers classified a lap or more down, which inflates every circuit's and constructor's DNF rate well above the true retirement rate (42.4% flagged vs. 14.6% true retirements dataset-wide). This required an explicit caveat wherever the metric is shown, rather than a silent redefinition of the field.

- **Small-sample correlations.** Circuit-level correlations (Section 0.9) are computed over only 31 circuits, which is a small sample; these are reported as suggestive patterns rather than statistically robust conclusions.

- **Overlapping data points.** With roughly 2,000+ driver–race records, plain scatter plots became unreadable overlapping clouds for both the Qualifying vs Race and Race Strategy modules; this was addressed by switching to jittering (small dataset) or 2D density heatmaps (larger dataset) rather than trying to force a scatter plot to scale.

- **Radar-chart legibility.** Radar charts degrade quickly past two or three overlapping series, which constrained the Driver Performance Explorer's radar view to at most two drivers at a time and kept it as a secondary, supporting chart rather than the primary comparison tool.

- **Cross-module integration.** Merging seven independently built modules into one dashboard surfaced callback conflicts and inconsistent styling that had to be resolved jointly, which took more coordination than any single module's chart-design decisions.

- **Filter-dependent KPIs.** Several summary labels (e.g. "Most Dominant," "Best Improver," "Most Reliable" in Constructor Evolution) are computed only over the currently selected constructors and season range, so the same team can hold or lose a title purely by widening a filter — a behavior that had to be documented rather than treated as a bug.

5. Conclusion:

- A single, centralized preprocessing and feature-engineering pipeline is what makes every dashboard module trustworthy and mutually consistent, rather than each module maintaining its own definitions.
- Qualifying position has a real but only moderate influence on race result (grid–finish correlation of 0.33); race-day performance can still overturn a poor grid slot, as shown by outlier gains of up to +18 positions.
- Constructor dominance changed hands mid-window: Mercedes led 2019–2021, Red Bull overtook from 2022, and the two finished 2024 within a few percentage points of each other in cumulative points, while McLaren's 2024 season was the single largest year-over-year improvement among the default constructors.
- Circuit design dominates lap-speed variation far more than season-to-season car development (an ∼89 km/h gap between the fastest and slowest circuits), while overtaking (position gain) is structurally close to zero at most circuits by construction of the fixed-size grid.
- Race outcomes are driven far more by pace than by pit-stop execution: pit duration and count show near-zero correlation with positions gained (−0.11 and −0.04 respectively), while pace rank correlates strongly with final position (0.83); a slow stop is a penalty to avoid, not a lever to gain positions.
- Chart-type selection was deliberate throughout every module — geographic data was mapped, accumulating data was charted as a line, categorical composition was shown as a stacked bar, and overlapping raw points were replaced with density heatmaps or jittering once volume made a plain scatter unreadable.

Future Work

- Extend the dataset backward before 2019 (or forward as new seasons are completed) to test whether the qualifying–finish correlation and pit-strategy findings hold across regulation eras, not only the current ground-effect/hybrid era.
- Add a true-retirement-only toggle to every DNF-rate view (circuit, constructor) so users can switch between the currently flagged rate and the stricter true-retirement definition without leaving the page.
- Incorporate weather and safety-car data per race, since both are known confounders for pit-stop timing and position-gain outcomes that the current dataset does not capture.
- Add a driver-vs-driver head-to-head export (PDF/CSV) from the Driver Performance Explorer, so a specific two-driver comparison can be saved directly from the dashboard rather than only viewed on screen.
- Deploy the dashboard publicly (current deployment link is a placeholder) and add authentication-free shareable URLs for individual filtered views.

6. Link to source code:

GitHub repository: https://github.com/sssinghm/CS661-F1-Insights

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